

M E R N Solution Architecture

shopEZ — B2C E-Commerce Marketplace (MERN Stack)

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PREPARED BY:	JAMPA KAILASH JOHNITH
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Solution Architecture

1. Layered System Architecture



FIGURE 9.1 — 4-LAYER SYSTEM ARCHITECTURE DIAGRAM

2. MVC Pattern Mapping

MVC Layer	MERN Component	Responsibilities
View	React.js (Frontend SPA)	Renders UI components; manages local state; dispatches API calls via Axios; updates global state via Context/Redux
Controller	Express.js Controllers	Receives validated HTTP requests; orchestrates business logic; invokes Model methods; formats and returns JSON responses
Model	Mongoose Schemas & Models	Defines data structure, validation rules, and type constraints; executes lifecycle hooks; interfaces with MongoDB Atlas

3. UML Class Diagram — Mongoose Models

```

places
1
0..*
owns
1
1
writes    1
0..*
receives
1
0..*
classifies
1
0..*
UserModel
+ObjectId _id
+String name
+String email
+String password
+String role
+Array addresses
+matchPassword(entered) : Boolean
+pre_save_hook() : void
ProductModel
+ObjectId _id
+String name
+Number price
+Number stock
+Number rating
+ObjectId category
+Array images
+Array reviews
OrderModel
+ObjectId _id
+ObjectId user
+Array orderItems
+Object shippingAddress
+Number totalPrice
+Boolean isPaid
+String orderStatus
CartModel
+ObjectId _id
+ObjectId user
+Array items
+Date updatedAt
CategoryModel
+ObjectId _id
+String name
+String slug
ReviewModel
+ObjectId _id
+ObjectId user
+ObjectId product
+Number rating
+String comment

```

FIGURE 9.2 — UML CLASS DIAGRAM: MONGOOSE DATA MODELS

4. Deployment Topology

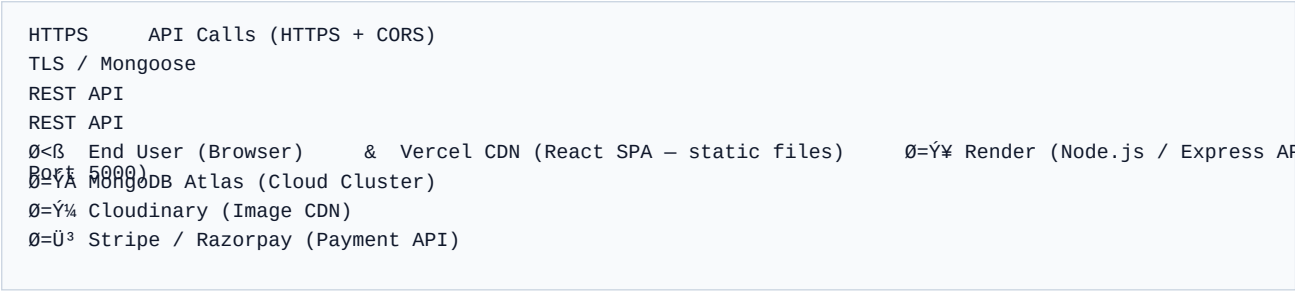


FIGURE 9.3 — CLOUD DEPLOYMENT TOPOLOGY

5. Deployment URL Reference

Component	Platform	URL
Frontend (React SPA)	Vercel	https://e-commerce-application-neon-five.vercel.app/
Backend (Node.js API)	Render	https://shopez-api-c30e.onrender.com
Database	MongoDB Atlas	Cloud-hosted (M0 Free Tier)
Image Storage	Cloudinary	res.cloudinary.com

